BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MALAKAND

INTERMEDIATE (12TH CLASS) EXAMINATION, 2025

BIOLOGY

Past Paper 2025 • Paper Code: 1

Retyped faithfully from the original board paper — same questions, no changes.

SECTION "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** 1

Note: There are Three Sections in this paper i.e. A, B and C. Attempt Section-A and return it to the Superintendent within the given time. No marks will be awarded for Cutting, Erasing and Overwriting. Mobile Phone is strictly prohibited in Examination Hall. Q.1. Choose the correct option and shade one of (A, B, C, D) in the given Bubble Answer Sheet.

Q1. Sense of smell is called:

(A) Gustation
(B) Olfaction
(C) Tactation
(D) None

Q2. Polydipsia,means:

(A) Large volume of sweating
(B) Excessive glucose in body
(C) Excessive hunger
(D) Excessive thirst

Q3. The volume of air exchanged during one breath in and out in quite breathing is called:

(A) Tidal volume
(B) Residual volume
(C) Lung volume
(D) Lung capacity

Q4. A single ova contain:

(A) XY chromosome
(B) Y chromosome
(C) X chromosome
(D) XXY chromosome

Q5. Which of the following carry genetic information?

(A) tRNA
(B) mRNA
(C) rRNA
(D) None

Q6. Hormones derived from cholesterol are called:

(A) Steroids
(B) Peptides
(C) Non-steroidal
(D) Protein hormones

Q7. Ozone layer is present in:

(A) Troposphere
(B) Mesosphere
(C) Stratosphere
(D) Thermosphere

Q8. Largest bone in human body is:

(A) Clavicle
(B) Patella
(C) Radius
(D) Femur

Q9. For detection of a gene in DNA library _____ is used:

(A) Primer
(B) Tag polymerase
(C) Probe
(D) Bloting

Q10. DNA and histones together form a bead like structure called:

(A) Mesosome
(B) Nucleosome
(C) Polysome
(D) Centrosome

Q11. The discontinuous growing daughter strand during DNA replication is called:

(A) Lagging strand
(B) Leading strand
(C) Smooth strand
(D) Continuous strand

Q12. Changes in gene frequencies in small population by chance is called:

(A) Gene pool
(B) Mutation
(C) Genetic change
(D) Genetic drift

Q13. _____ prepare body for stress and emergency situation.

(A) Adrenaline
(B) Thyroxin
(C) Insulin
(D) Nor Adrenaline

Q14. An individual with similar alleles is called:

(A) Homozygous
(B) Heterozygous
(C) Monoicious
(D) Diecious

Q15. The optimum temperature for sperm development is about:

(A) 37 Cº
(B) 36 Cº
(C) 35 Cº
(D) 33 Cº

Q16. Who invented the PCR technique?

(A) Sanger
(B) Cohn
(C) Boyer
(D) Karry Mullis

Q17. Learning Behaviour will be:

(A) Adoptable
(B) Extrinsic
(C) Progressive
(D) All of these

Q18. Hunger center is located in:

(A) Hypothalamus
(B) Cerebellum
(C) Medulla
(D) Pones

SECTION "B"

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40

Q2. Give short answers to any TEN (10) questions. All questions carry equal marks.

i. Describe the structure of taste receptors.

ii. What is RNA splicing?

iii. Define Co-dominance and over dominance with example.

iv. What is the role of restriction endonucleases in biotechnology?

v. Describe the structure of bone.

vi. What is pneumonia? Write down its symptoms and treatment.

vii. Write short note on kidney transplantation.

viii. What is primary and secondary succession?

ix. Define neuron. What are the different types of neurons?

x. Shortly describe Genetic code.

xi. Define struggle for existence and variation.

xii. What are the problems related to rapid growth in human population?

xiii. Shortly describe spermatogenesis.

SECTION "C"

Maximum Marks: 27

Note: Attempt any THREE (03) questions from the following. Each question carries equal marks.

Q3. (Marks 09) Write detail note on peripheral nervous system.

Q4. (Marks 09) Write brief note on events of DNA replication.

Q5. (Marks 09) Briefly explain the structure and function of human kidney.

Q6. (Marks 09) Write note on:

a) [05] Transport of O_2 as oxyhemoglobin

b) [04] Ecological Pyramids